

# History of Celestial Mechanics

From Ptolemy to Satellite Constellations

Press → to advance

# Historical Timeline



# The Ptolemaic Universe

Image  
placeholder:  
Ptolemaic  
geocentric  
model

`images/ptolemaic_system.jpg`

- ▶ Earth at the center of the universe
- ▶ Planets move on **epicycles** (circles upon circles)
- ▶ Explained retrograde motion
- ▶ Dominated astronomy for **1400 years**



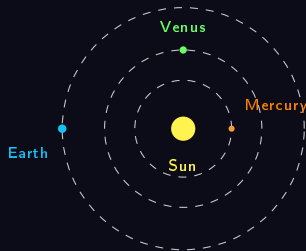
# Copernicus: The Heliocentric Revolution

Portrait  
placeholder:

Nicolaus  
Copernicus  
(1473–1543)

`images/copernicus.jpg`

- ▶ Sun at the center
- ▶ Simpler explanation of retrograde motion
- ▶ *De Revolutionibus* (1543)
- ▶ Still used circular orbits!



# What is Retrograde Motion?

- ▶ Planets sometimes appear to move **backward** against the background stars
- ▶ The word *planet* comes from the Greek *planētēs* — “**wanderer**”
- ▶ Caused by Earth overtaking (or being overtaken by) another planet in its orbit

Apparent path of Mars on the sky



# Animation: Retrograde Motion



Retrograde Motion

Click to play

## Key Insight

Both geocentric and heliocentric models explain retrograde motion. But the heliocentric model is far simpler — Occam's razor at work.

# Epicycles = Fourier Series

Any closed curve can be decomposed into circular motions:

$$z(\theta) = \sum_{n=-N}^N c_n e^{i n \theta}$$

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Ptolemy's epicycles are a truncated Fourier series!

# Animation: Epicycles & Fourier Reconstruction



Ptolemy's Epicycles

## Key Insight

Ptolemy's epicycles are a truncated Fourier series.  
Any periodic motion can be decomposed into a sum  
of uniform circular motions.

# Galileo: Projectile Motion

Horizontal and vertical motions are **independent**:

$$x(t) = v_0 \cos \theta \cdot t$$

$$y(t) = v_0 \sin \theta \cdot t - \frac{1}{2} g t^2$$

Eliminate  $t$ :

$$y = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

⇒ This is a **parabola**!

Maximum range at  $\theta = 45^\circ$ :

$$R_{\max} = \frac{v_0^2}{g}$$

# Animation: Projectile Trajectories



Galileo: Projectile Motion

Click to play

## Key Insight

Horizontal and vertical motions are independent.  
This single principle gives us the parabolic trajectory.

# Kepler's Three Laws

**Law 1** — Orbits are **ellipses**, Sun at one focus:

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

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**Law 3** — Orbital period vs. semi-major axis:

$$T^2 = \frac{4\pi^2}{GM} a^3$$

# Animation: Kepler's Laws



Kepler's Laws

Click to play

## Key Insight

Kepler's laws are purely empirical — he found patterns in data, but had no explanation for *why* orbits are ellipses.

# Newton's Universal Gravitation

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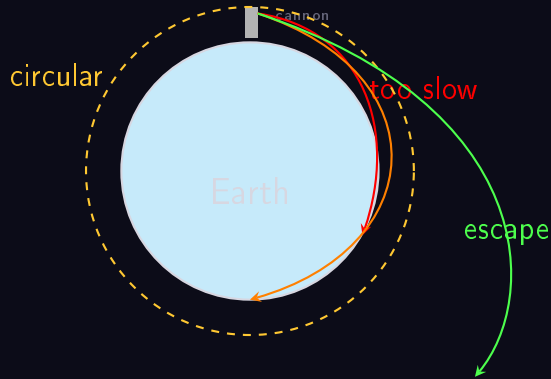
$$\ddot{\vec{r}} = -\frac{GM}{|\vec{r}|^3} \vec{r}$$

**Binet equation** (polar coordinates,  $u = 1/r$ ):

$$\frac{d^2 u}{d\theta^2} + u = \frac{GM}{L^2}$$

Solution: **conic sections** — ellipse, parabola, or hyperbola.

# Newton's Cannon: From Fall to Orbit



# Animation: Newton's Cannon



Newton's Cannon

Click to play

## Key Insight

The same force that drops an apple curves a cannonball into orbit. Newton unified terrestrial and celestial mechanics.

# Interactive Demo: Orbital Simulator

Live orbital mechanics simulation



Open Orbital Simulator

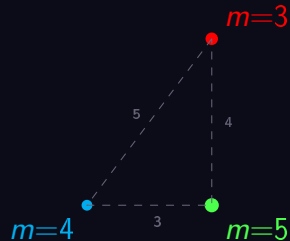
[skychart.astrageek.ru/astra-etudes/orbital-simulator](https://skychart.astrageek.ru/astra-etudes/orbital-simulator)

Scan QR or visit URL in your browser

# The Three-Body Problem

- ▶ Two bodies → **exact solution**  
(Kepler orbits)
- ▶ Three bodies → **no general closed-form solution**
- ▶ Poincaré (1890): proof of **chaotic behavior**
- ▶ Extreme sensitivity to initial conditions

## Pythagorean problem



Masses at rest in a  
3-4-5 right triangle

# Animation: Three-Body Problem



Three-Body Problem

Click to play

## Key Insight

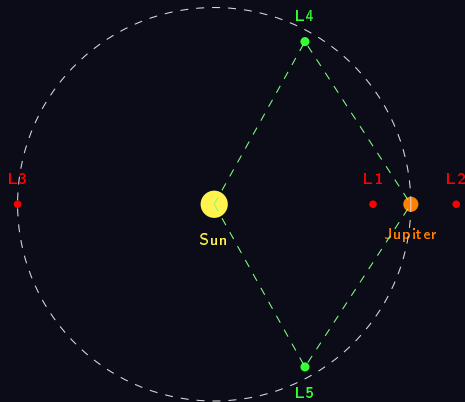
Adding just one more body destroys exact solvability. The universe is fundamentally chaotic at the three-body level.

# Lagrange Points

In the **rotating frame** of  
Sun–Planet:

- ▶ 5 equilibrium points (L1–L5)
- ▶ L1, L2, L3: collinear, **unstable**
- ▶ L4, L5: triangular, **stable**

Jupiter's L4 and L5 host **Trojan asteroids**  
(>12 000 known objects!)



# Animation: Lagrange Points & System Rotation



Lagrange Points

## Key Insight

Equilibrium exists even in a chaotic gravitational landscape. L4 and L5 are stable — home to thousands of Trojan asteroids.

# From Newton to Einstein

Newton's Binet equation:

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This causes **perihelion precession**:

$$\Delta\varphi = \frac{6\pi GM}{c^2 a (1 - e^2)} \quad \text{per orbit}$$

# Animation: Mercury Perihelion Precession



Mercury Precession (GR)

Click to play

## Key Insight

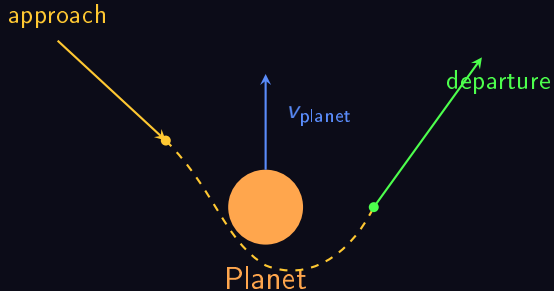
Newton was “wrong” — but only by 43 arcseconds per century. Einstein’s tiny correction revealed the curvature of spacetime.

# Modern Celestial Mechanics

- ▶ **JWST** at Sun–Earth L2 (Lagrange, 1772)
- ▶ **GPS** requires GR time corrections of  $\sim 38 \mu\text{s}/\text{day}$  (Einstein, 1915)
- ▶ **Voyager** gravity assists — slingshot past Jupiter and Saturn (Newton, 1687)
- ▶ **Starlink / OneWeb** constellations — thousands of satellites in low Earth orbit (Kepler, 1619)

Every modern space mission stands on 2000 years of celestial mechanics.

# Gravity Assists: The Cosmic Slingshot



In the **planet's rest frame**, speed is unchanged — the spacecraft merely deflects.

In the **Sun's frame**, the spacecraft gains up to:

$$\Delta v = 2 v_{\text{planet}}$$

(maximum gain for a  $180^\circ$  deflection)

Voyager 2 used four gravity assists to reach interstellar space.

# Summary

|            |                                          |
|------------|------------------------------------------|
| Ptolemy    | Epicycles (= Fourier series!)            |
| Copernicus | Heliocentric model                       |
| Galileo    | Projectile motion = parabola             |
| Kepler     | Elliptical orbits, three laws            |
| Newton     | Universal gravitation, $F \propto 1/r^2$ |
| Lagrange   | Equilibrium points L1–L5                 |
| Einstein   | GR correction $\rightarrow$ precession   |

From circles to spacetime curvature —

# Thank you!

Questions?